

Experimental Evaluation of Dynamic-Fee Hooks

(section source for the ICBC paper)

I. EXPERIMENTAL EVALUATION

The framework’s evaluation layer runs every fee policy and a family of static baselines over identical replayed market histories and scores the liquidity provider (LP) against simply holding the deposited tokens. The experiment plays two roles: it exercises the full configure–deploy–replay path at scale, and it provides a controlled case study of whether the supported dynamic policies improve LP outcomes over a static fee under the declared market model.

A. Evaluated Policies

All policies share the fee box $[1, 100]$ bps (fees are `uint24` values in hundredths of a basis point), and every updated fee is clamped to that box (left implicit below). Directional fees are written $f^{0 \rightarrow 1}$ (selling token0) and $f^{1 \rightarrow 0}$; every fee starts at 30 bps unless noted. Gas is the EVM-measured median per swap, including the 21k intrinsic transaction cost. Table I summarizes each policy’s signal, update rule, and concrete parameters.

DAHook, ABHook, and MEVChargeHook are endogenous (their oracle arguments are unused); BAHook and VolatilityHook read two Chainlink-compatible feeds, which is what their higher gas reflects.

B. Simulated Market and Reproducibility

One *cell* of the experiment’s factorial matrix, a combination of fee policy, trading pair, day-window, and gas-price scenario, replays a *window* of 1,440 one-minute Binance candles from 2024 against a Uniswap v4 pool holding a 20M USDT basket, split equally by value into a full-range position. The pool opens at the external market price of the first candle; a mock Chainlink-compatible feed tracks each candle’s close; the first 60 candles execute but are excluded from measurement. Three pairs are studied: ETH/SHIB and ETH/USDC (volatile) and USDC/USDT (stable). Windows are stratified by realized volatility into terciles (24 calm, 24 medium, and 24 stormy days per pair), so market regimes are represented by design.

Every cell is identified by the manifest of Eq. (7) (configured contract’s source hash, initial price and liquidity, a fingerprint of the window’s own candles, and the RNG seed), so any change to a contract or input identifies exactly the affected cells. Because every swap executes in a real EVM (Foundry), gas is *measured* rather than assumed: a calibration run records each policy’s median gas per swap (76k for the static hook, 88–124k for the adaptive ones), and these measured figures feed back into both traders’ decisions below. Fees follow v4

accounting exactly: they accrue to the position and do not compound.

C. Two-Flow Trader Model

Following the standard microstructure framing, order flow is split into an informed arbitrageur and uninformed users (UU).

1) *Arbitrageur*: With fee fraction f and $\gamma = 1 - f$, the pool price may sit anywhere inside the no-arbitrage band $[P\gamma, P/\gamma]$ around the external price P . Once per candle, after the retail flow, the arbitrageur moves it to the nearest band edge if profitable *after gas*: with pool liquidity L , sqrt-price s and target t , the closed-form profit

$$\Pi_{0 \rightarrow 1}^* = \frac{L(s - t)^2}{s}, \quad \Pi_{1 \rightarrow 0}^* = \frac{L(t - s)^2}{\gamma s}, \quad (1)$$

(derived in Sec. II) must exceed $g p_{\text{gas}}$, where g is the policy’s measured median gas and p_{gas} the scenario’s gas price (5, 20, or 80 gwei), valued at the ETH price series. For the size-dependent MEVChargeHook the optimum is found by ternary search; profit is unimodal in size. Π^* is a decision quantity only: all reported metrics are computed from the balance changes of executed swaps.

2) *Uninformed users*: UU behavior follows the acceptance model of our prior work [1]: a candidate trade evaluates the normalized outcome ratio

$$r = \frac{V_{\text{out}} - V_{\text{in}} - G}{V_{\text{out}} + V_{\text{in}} + G}, \quad (2)$$

where V_{in} and V_{out} are the two legs valued at external prices, so fee *and* slippage are inside r , and G is the trade’s own gas. Execution is probabilistic,

$$P_{\text{exec}} = \begin{cases} 1, & r \geq 0, \\ e^{-\lambda|r|}, & r < 0, \end{cases} \quad (3)$$

with $\lambda = \ln 2 / |r(30 \text{ bps})| \approx 461.4$, so that half of the willing flow declines at a 30 bps total transaction cost. This calibration is what makes participation fee-sensitive: at these trade sizes, the raw exponent $e^{-|r|}$ is nearly flat across the fee box. The value of λ is a modeling assumption rather than a measured quantity and is varied in sensitivity analysis.

Each minute c , each direction independently draws one candidate trade of size

$$q(c) = \frac{v_0(c)}{2} e^{\sigma Z - \sigma^2/2}, \quad Z \sim \mathcal{N}(0, 1), \quad \sigma = 1, \quad (4)$$

where $v_0(c)$ is the minute’s potential demand (Eq. (6) below), split equally between the two directions. The lognormal multiplier is mean-preserving ($\mathbb{E}[e^{\sigma Z - \sigma^2/2}] = 1$), so randomness

TABLE I
THE EVALUATED FEE POLICIES.

Policy	Update	Rule	Parameters	Gas
Static	never	constant fee, both directions	$f \in \{5, 30, 60, 100\}$ bps; 30 bps = baseline	76–77k
DAHook	every trade	after a swap in direction d : $f^d \leftarrow f^d + \delta$, $f^{-d} \leftarrow f^{-d} - \delta$	$\delta = 1$ bps	88k
BAHook	once per block	external ratio $\rho_b = p_{0,b}/p_{1,b}$ from two feeds: $f^{0 \rightarrow 1} \leftarrow f^{0 \rightarrow 1} + \delta \operatorname{sgn}(\rho_{b-1} - \rho_b)$, opposite side mirrored	$\delta = 5$ bps	108k
ABHook	every trade	$\Delta_t = \text{relative } \sqrt{\text{price}} \text{ move of the pool itself: } f^{0 \rightarrow 1} \leftarrow f^{0 \rightarrow 1} - c \Delta_t, f^{1 \rightarrow 0} = K - f^{0 \rightarrow 1}$	$c = 100$ bps per unit move; $K = 60$ bps (each side in $[1, 59]$ bps)	94k
VolatilityHook	once per block	$f = f_{\text{base}} + c \hat{\sigma}$, both directions; $\hat{\sigma} = \text{EMA per-minute st. dev. of the external ratio's relative change}$	$f_{\text{base}} = 20$ bps, $c = 2 \times 10^6$, $\alpha = 0.1$	123k
MEVChargeHook	every trade	30 bps base + surcharges in trade size and for repeated senders	cooldown 15 s; range $[30, 100]$ bps	95k

produces realistic heavy-tailed trade sizes without distorting the calibrated volume; the trade then executes all-or-nothing with probability P_{exec} . Trades are discrete because they must displace the pool price to create arbitrage opportunities, and each one carries its own gas cost; all draws are pre-generated from a recorded seed, so the randomness stays reproducible. One candidate trade per direction per minute is the coarsest discretization of the measured volume.

3) *Demand calibration*: The minute-level potential demand $v_0(c)$ combines a measured intraday shape with one assumed scale parameter, κ . With $V_0(c), V_1(c)$ the 2024 per-minute Binance quote volumes of the pair's two legs, the intraday shape is

$$\hat{s}(c) = \frac{g(c)}{\sum_{c' \in \text{day}} g(c')}, \quad g(c) = \sqrt{V_0(c) V_1(c)}, \quad (5)$$

where the bar averages over the days of 2024, so $\hat{s}$ sums to one over an average day, and

$$v_0(c) = \kappa \cdot \text{basket} \cdot \hat{s}(c), \quad (6)$$

with the basket the pool's initial 20M USDT deposit. Hence κ is the pool's daily turnover relative to its own TVL. The reference point $\kappa = 1$ is fixed by the parameterization, not by any result. The experiment is repeated at $\kappa \in \{0.03, 0.1, 0.3, 1, 3\}$, each a full matrix never pooled with another, which moves the measured arbitrage share of volume across 89%, 58%, 25%, 13%, 16% (Table ??, Fig. 1).

D. Design and Statistics

One operating point is a full factorial over 9 policy configurations (the five hook policies plus static baselines at 5/30/60/100 bps), times 3 pairs, 72 windows, and 3 gas scenarios: 5,832 cells. Five operating points give 29,160 cells. Figure 2 illustrates the policies' qualitative behavior on one high-volatility day: directional policies widen the fee against the market and narrow the opposite side; the constant-sum policy shifts little; *VolatilityHook* rises to the fee ceiling as volatility increases; *MEVChargeHook* alternates between

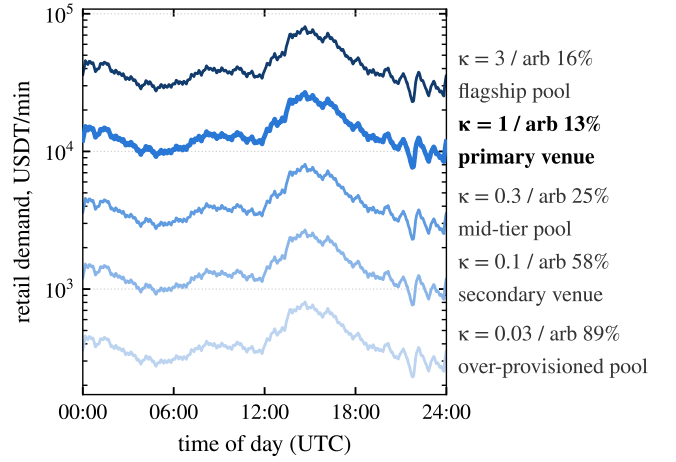


Fig. 1. Demand calibration, Eq. (6): the intraday shape $\hat{s}(c)$ is measured (median 2024 day of the pair's Binance legs); the single parameter κ scales its height, which determines the measured arbitrage share of volume.

its base and capped rate. The LP outcome of a cell is $\text{net} = R_f - IL = -L_{\text{net}}$, the negative of the fee-adjusted loss of Eq. (4): the LP's position and accrued fees against simply holding the deposit, all marked at end-of-window prices.

Every comparison is *paired*: a policy's window result is differenced against the pre-declared 30 bps static baseline on the same window, pair, and gas price, removing the day-to-day variation that dwarfs policy effects. Effects are reported as the median paired difference with a 95% percentile-bootstrap confidence interval (10,000 seeded resamples), written median [lower, upper] throughout; a comparison counts as a win or loss only if the interval excludes zero. Headline numbers aggregate cell-first (one median per pair $\times$ regime $\times$ gas cell, then across cells), so three gas scenarios of one day are never treated as independent observations. A post-hoc comparison against the best static fee in the box (60 bps) is reported and labeled exploratory.

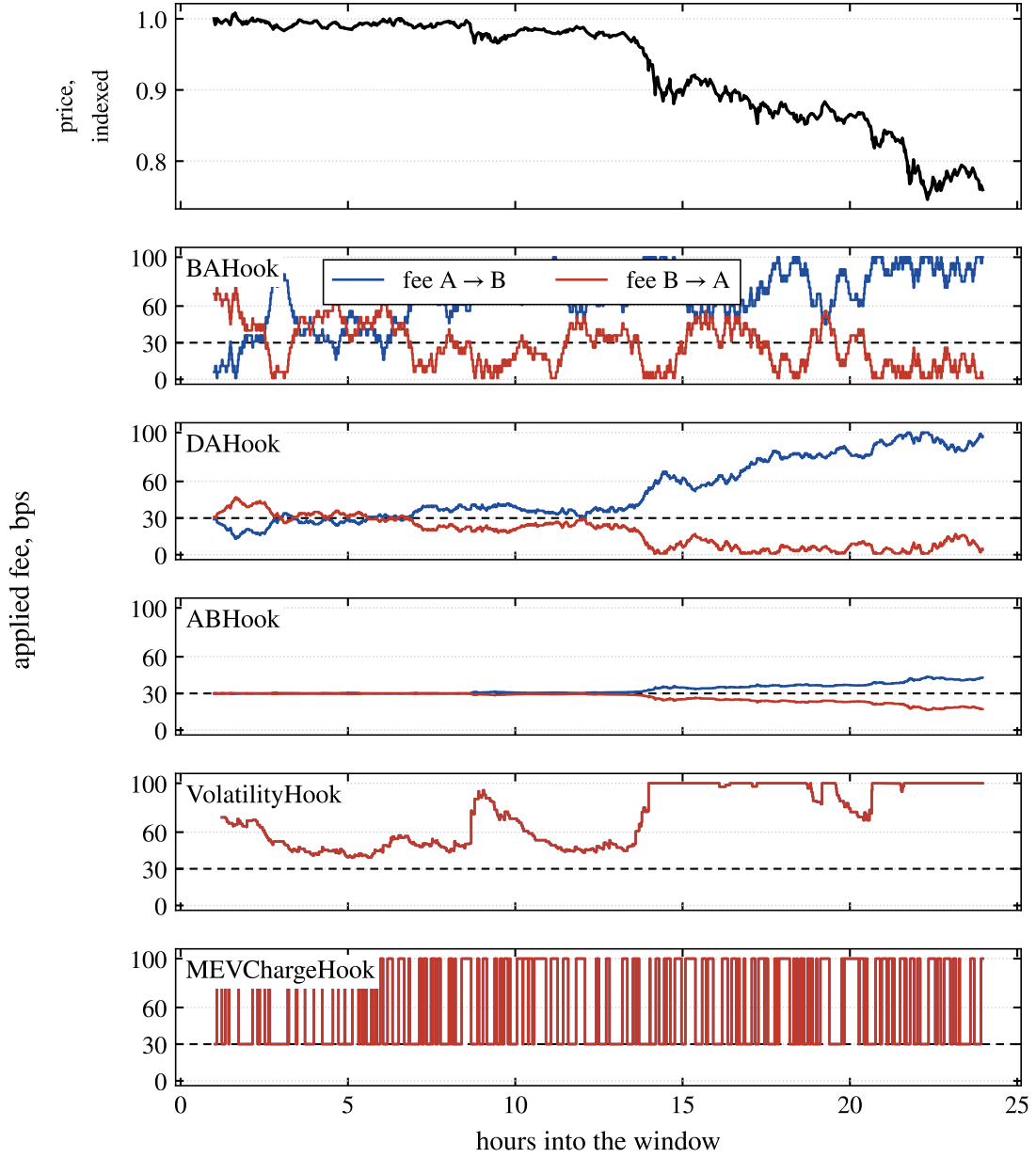


Fig. 2. Fees charged by each policy over one high-volatility ETH/SHIB day (2024-03-01, 20 gwei, $\kappa=1$). Top: the external price, indexed to the window open. Below, one panel per policy: the fee state per direction on the minute grid (the quoted fee persists between swaps), blue for token0→token1, red for the reverse; symmetric policies show a single curve. Dashed = the 30 bps baseline; the fee box caps all policies at 100 bps.

TABLE II
MEDIAN PAIRED ADVANTAGE OVER THE 30 BPS STATIC BASELINE AT THE REFERENCE POINT ($\kappa=1$): MEDIAN ACROSS THE 18 VOLATILE-PAIR STRATA, 95% BOOTSTRAP CI. NEGATIVE = WORSE THAN THE BASELINE.

Policy	advantage vs 30 bps, USDT/window
Static 60 bps	-2 [-186, +266]
BAHook	-100 [-453, +350]
DAHook	-235 [-381, -104]
ABHook	-460 [-630, -288]
VolatilityHook	-752 [-1069, -280]
MEVChargeHook	-913 [-1659, -728]

E. Results

1) *Aggregate performance:* Table II reports the primary comparison at the reference point: no evaluated dynamic

policy beats the static 30 bps baseline; four lose decisively and BAHook is indistinguishable from it. The same holds at every other operating point: pooled across regimes and gas scenarios, no hook’s interval excludes zero at any κ . Among the static levels themselves, both boundary fees lose to 30 bps decisively while 60 bps does not, a plateau at 30–60 bps consistent with the model’s predicted revenue-maximizing fee of $f^* \approx 2/\lambda \approx 43$ bps.

Decomposing the paired differences shows that the impermanent-loss term is policy-invariant: it differs by single-digit USDT across policies, against fee-income effects in the hundreds. Arbitrage pins the end-of-window pool price to the

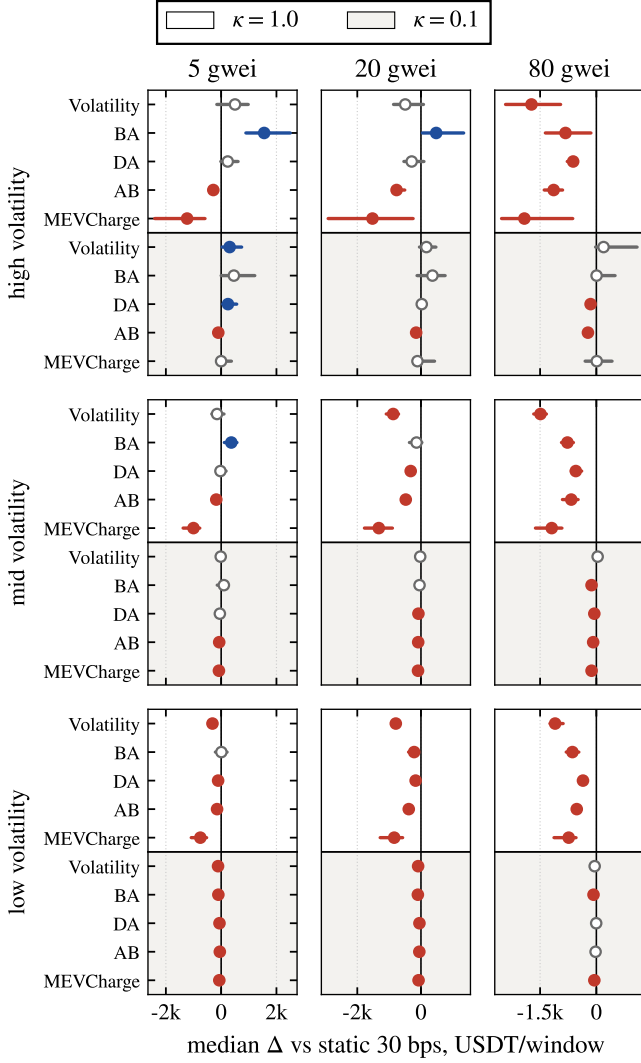


Fig. 3. Conditional structure by volatility regime $\times$ gas scenario (volatile pairs pooled, 48 windows per panel). Each cell is split top/bottom: the reference point ($\kappa=1$, white) above the informed-heavy point ($\kappa=0.1$, arbitrage 58% of volume, shaded), same hook order and x-axis in both halves. Filled marker = 95% CI excludes zero; blue wins, red loses, open grey = interval covers zero.

external price under every policy, so adaptation cannot reduce adverse selection in this model; it can only price the flow, and it prices it slightly worse than a well-chosen constant while paying its own gas premium on top.

2) *Conditional performance*: Splitting results by both experimental axes together shows a consistent pattern (Fig. 3): every policy’s advantage over the baseline improves monotonically with volatility and degrades monotonically with gas, in every regime and at every κ . At the reference point ($\kappa=1$), **BAHook** beats the baseline by +1,556 [+894, +2,501] USDT per window in high-volatility windows at 5 gwei, but loses or ties at 80 gwei. Which policy carries this advantage shifts with the flow mix (Fig. 5): **BAHook** at retail-dominated κ , **VolatilityHook** and **DAHook** at informed-heavy κ (Fig. 3, bottom half of each cell).

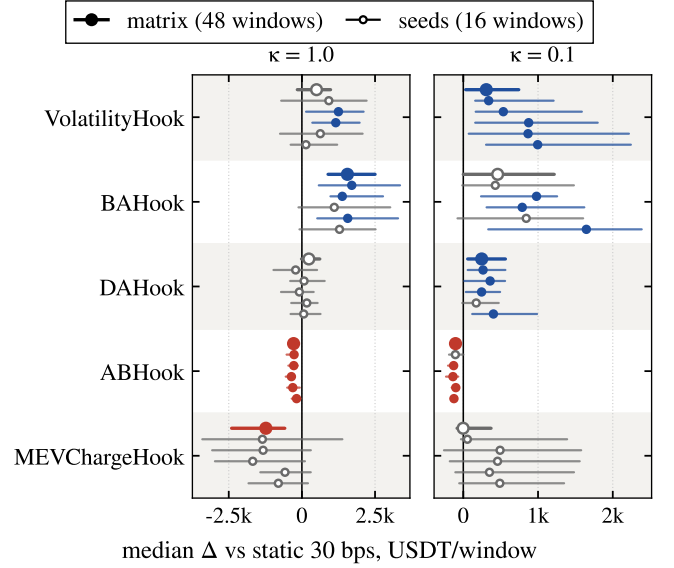


Fig. 4. Seed robustness of the conditional estimates (high-volatility windows, 5 gwei, volatile pairs pooled): for each policy, the matrix-of-record estimate (bold, 48 paired windows) above its five seed replicates (16 windows each). Filled marker = 95% bootstrap interval excludes zero. Replicate intervals are wider only because the replicate window set is three times smaller; the sign is unanimous across seeds for every reported finding.

These conditional findings are stable across random seeds: a five-seed replication of a 24-window subset preserves the sign of every conditional finding (Fig. 4), and **ABHook** stays significantly negative at nearly every seed, showing the procedure also detects real disadvantages.

3) *Gas overhead*: The gas gradient follows directly from the policies’ own overhead. Oracle-reading hooks cost 108–124k gas per swap against the static hook’s 76k, and that difference enters retail’s acceptance test through G in Eq. (2). At 80 gwei even the static hook’s $\sim$ \\$19 per swap amounts to $\sim$ 47 bps of a typical \\$4,000 retail trade, so transaction costs rival the fee box itself. The informed share of volume rises with gas accordingly, from 12.4% to 13.7% across 5 $\rightarrow$ 80 gwei at $\kappa=1$, as small retail trades are priced out first.

Overall, none of the evaluated dynamic policies outperforms a well-chosen static fee unconditionally. A reproducible conditional advantage exists in high-volatility windows at low gas prices; which policy carries it depends on the informed share of flow, and the advantage is consumed by the policies’ own gas premium as gas rises. For the framework, what matters here is that this conditional structure could be measured at all: paired, seed-checked, and decomposed along pre-declared axes, with every policy running through the same implementation path.

F. Threats to Validity

The declared model limits: retail has no strategy and no correlated demand; one arbitrageur acts once per candle with perfect information and no mempool competition; liquidity is passive and full-range; mock feeds validate integration, not live oracle behavior; the retail scale κ is swept over

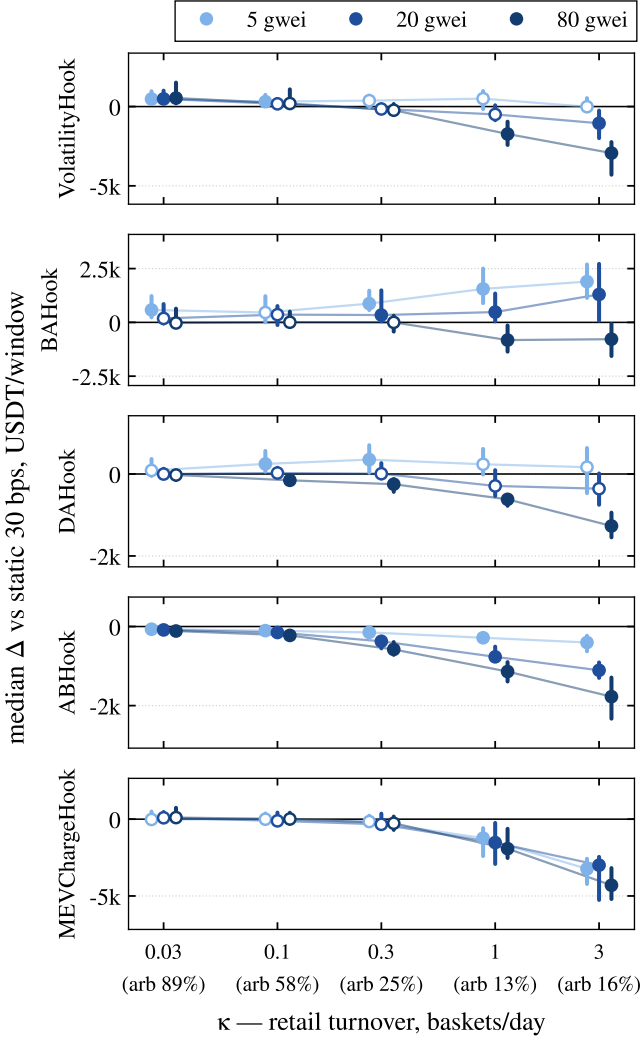


Fig. 5. The high-volatility slice across the retail scale: advantage over the baseline, by κ and gas scenario.

two orders of magnitude, not calibrated to real pools; retail lumpiness is fixed at one trade per direction per minute; results cover one year, three pairs, and one venue's data. The main matrix uses one recorded seed; the five-seed replication of Fig. 4 covers a window subset rather than the full matrix. The conditional corner finding was not pre-registered and is reported as observed structure; the unconditional findings and the failed-and-reported gas prediction follow the pre-registered protocol.

II. DERIVATION OF THE ARBITRAGE OPTIMUM

Within one tick range a Uniswap v3/v4 pool with liquidity L obeys the constant-product amount relations in sqrt-price coordinates: moving the pool price from $\sqrt{p} = s_1$ to s_2 exchanges

$$\Delta y = L(s_2 - s_1), \quad \Delta x = L\left(\frac{1}{s_1} - \frac{1}{s_2}\right), \quad (7)$$

and the LP fee f ($\gamma = 1 - f$) is charged on the input amount before it enters the curve.

1) *Band edges*: Selling token0 (zero-for-one) at pool sqrt-price s realizes a marginal price γs^2 in token1 per token0; the trade is profitable against the external price P while $\gamma s^2 > P$, i.e. while $s > \sqrt{P/\gamma} = t_{0 \rightarrow 1}$. Buying token0 costs s^2/γ marginally and is profitable while $s^2/\gamma < P$, i.e. while $s < \sqrt{P\gamma} = t_{1 \rightarrow 0}$. Hence the no-arbitrage band $[P\gamma, P/\gamma]$ on the pool price, and the profit-maximizing trade stops exactly at the near edge: the integrand of marginal profit changes sign there, so profit is unimodal in trade size with its maximum at the edge.

2) *Sell side*: Moving the pool from s down to $t = t_{0 \rightarrow 1}$ requires a net token0 input $\Delta x = L(1/t - 1/s)$, for which the trader pays the grossed-up $\Delta x/\gamma$, and yields $\Delta y = L(s - t)$ of token1. Valued at the external price $P = \gamma t^2$,

$$\begin{aligned} \Pi_{0 \rightarrow 1} &= L(s - t) - \frac{P}{\gamma} L\left(\frac{1}{t} - \frac{1}{s}\right) = L(s - t) - L t^2 \frac{s - t}{ts} \\ &= L(s - t) \left(1 - \frac{t}{s}\right) = \frac{L(s - t)^2}{s}. \end{aligned} \quad (8)$$

3) *Buy side*: Moving the pool from s up to $t = t_{1 \rightarrow 0}$ requires net token1 input $L(t - s)$, paid as $L(t - s)/\gamma$, and yields $\Delta x = L(1/s - 1/t)$ of token0 worth $P \Delta x$. With $P = t^2/\gamma$,

$$\begin{aligned} \Pi_{1 \rightarrow 0} &= \frac{t^2}{\gamma} L \frac{t - s}{ts} - \frac{L(t - s)}{\gamma} = \frac{L(t - s)}{\gamma} \left(\frac{t}{s} - 1\right) \\ &= \frac{L(t - s)^2}{\gamma s}. \end{aligned} \quad (9)$$

The extra $1/\gamma$ relative to Eq. (8) arises because the fee is charged in token1, the numeraire the profit is measured in, so, unlike the sell side, it does not cancel against the external valuation of the input.

4) *Gas*: Gas adds a size-independent cost $g p_{\text{gas}}$, which shifts the profit curve down without moving its maximum; checking $\Pi^* > g p_{\text{gas}}$ after the size optimization is therefore equivalent to including gas inside it (the arbitrageur cannot split a trade across transactions in this model). For a size-dependent fee the closed form is replaced by a ternary search, valid because an input fee non-decreasing in size preserves unimodality.

REFERENCES

- [1] Irina Lebedeva, Dmitrii Umnov, Yury Yanovich, Ignat Melnikov, and George Ovchinnikov. Dynamic fee for reducing impermanent loss in decentralized exchanges, 2025.